INCH-POUND
MIL-DTL-24643/43E
22 August 2002
SUPERSEDING
MIL-C-24643/43D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS TYPES, LS2SJ, LS3SJ AND LS4SJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG sizes 14 and smaller. Uncoated for AWG sizes 12, class B stranding, ASTM B8, and larger.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation. (see table I for dimensions). Standard identification code applied by method 3.
Fourth -	Two, three or four conductors, as required, cabled together with fillers to form a firm, well rounded assembly.
Fifth -	Braided shield of tin coated copper.
Sixth -	Binder tape, applied helically with overlap.
Seventh -	Cross-linked polyolefin jacket. (Minimum average wall thickness 0.035 inch.)

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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TABLE I. Details.

Military part no. M24643/43	Conductors				Insulation wall thickness (min avg) (inch)	Lay of conductor maximum (inches)	Type's overall diameter (inches)		Conductor resistance (ohms)
	Size	No.	Size [1/]				LSSJ		
			ASTM B 286	ASTM B 8			min	max	
-01UO	22	2	22-19		0.016	1-1/2	0.261	0.275	16.54
-02UO	20	2	20-19		0.016	1-1/2	0.273	0.290	10.16
-03UO	18	2	18-19		0.016	1-1/2	0.295	0.310	6.47
-04UO	16	2	16-19		0.016	2-1/2	0.309	0.325	5.02
-05UO	14	2	14-19		0.016	2-1/2	0.337	0.350	3.17
-06UO	12	2		12(class B)	0.024	3	0.417	0.430	1.72
-07UO	11	2		10(class B)	0.025	3	0.447	0.460	1.085
-08UO	9	2		9(class B)	0.031	4	0.525	0.545	0.86
-09UO	7	2		7(class B)	0.032	4	0.600	0.615	0.54
-10UO	22	3	22-19		0.016	1-1/2	0.271	0.285	16.54
-11UO	20	3	20-19		0.016	1-1/2	0.284	0.300	10.16
-12UO	18	3	18-19		0.016	1-1/2	0.308	0.325	6.47
-13UO	16	3	16-19		0.016	2-1/2	0.323	0.340	5.02
-14UO	14	3	14-19		0.016	2-1/2	0.353	0.370	3.17
-15UO	12	3		12(class B)	0.024	3	0.440	0.455	1.72
-16UO	9	3		9(class B)	0.031	4	0.594	0.620	0.86
-17UO	20	4	20-19		0.016	1-1/2	0.303	0.320	10.16
-18UO	16	4	16-19		0.016	2-1/2	0.346	0.360	5.02
-19UO	14	4	14-19		0.016	3	0.380	0.395	3.17

1/ Size designation does not correlate to conductor size for all types and sizes.

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - omhs/1000 feet at 25°C, maximum .....(see table I)
Voltage withstand - volts, root mean square, minimum
Conductor to conductor 1200
Conductor to shield 600
Insulation resistance - Megohms -1000 feet minimum.
Conductor to shield 500
Jacket flaws ..... No failure
Conductor and shield continuity ..... No failure

Group A:
Visual and dimensional ..... No failure

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Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50
  - Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)
- Insulation
  - Ethylene propylene rubber
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2, minimum 1800
    - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals (aged)
- Insulation
  - Ethylene propylene
    - Air Oven
      - Tensile strength - percent of unaged, minimum 75
      - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
    - Air Oven
      - Tensile strength - percent of unaged, minimum 80
      - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125

- Shield - conformance to material, construction and coverage ..... No failure
- Surface transfer impedance
  - Milliohms per meter, maximum 700
  - EMP response time - dB, minimum 60

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Group D:

Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-041)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA - CC

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